REAL ANALYSIS II · SEMESTER 5 · MODULE 1

Answer key: Continuous Functions

Formal solutions to all 40 questions of the model paper.

BEFORE YOU READ AN ANSWER

Each answer starts with **Know first**: topic codes from File 1. If one feels new, read it first, try the question yourself, then read the solution.

Grey "Marks" notes show a typical split (a guide, not an official scheme): statement · set-up · key step · conclusion.

Section A · Short answers

3 marks each.

1 PYQ MTS-22 Q1

Continuity at c; constant function.

KNOW FIRST

T1 The ε–δ definition P2 Modulus (absolute value)

Let $A \subseteq \mathbb{R}$, $f : A \to \mathbb{R}$, $c \in A$. f is **continuous at c** if, given any $\varepsilon > 0$, there exists $\delta > 0$ such that if x is any point of A with $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.

Let $f(x) = b$ for all x. For any c and any ε > 0, take δ = 1. Then for every x, $|f(x) - f(c)| = |b - b| = 0 < \varepsilon$. So f is continuous at every c, i.e. on $\mathbb{R}$.

Marks: definition 1.5 · proof 1.5.

2

Neighbourhood form; limit form; isolated points.

KNOW FIRST

T2 Neighbourhoods, limits and isolated points P9 Limits of functions

Neighbourhood form (5.1.2). f is continuous at c iff for every ε-neighbourhood $V_\varepsilon(f(c))$ there is a δ-neighbourhood $V_\delta(c)$ with $f(A \cap V_\delta(c)) \subseteq V_\varepsilon(f(c))$.

Limit form. If c is a cluster point of A, f is continuous at c iff $f(c)$ is defined, $\lim_{x \to c} f(x)$ exists, and they are equal.

Isolated point. If c is not a cluster point, some $V_\delta(c)$ meets A only in {c}; then for $x \in A$ with $|x - c| < \delta$ we have x = c and $|f(x) - f(c)| = 0 < \varepsilon$. (E.g. $A = [0,1] \cup \{2\}$, c = 2, δ = ½.)

Marks: each part 1.

3 PYQ MTS-26 Q1

Discontinuity Criterion; [x] at 1.

KNOW FIRST

T3 The sequential criterion P6 Sequences and their limit theorems

Discontinuity Criterion (5.1.4). Let $f : A \to \mathbb{R}$, $c \in A$. f is discontinuous at c iff there is a sequence (x_n) in A with $x_n \to c$ but $(f(x_n))$ not converging to f(c).

Take $x_n = 1 - \dfrac{1}{n+1} \in [\dfrac{1}{2}, 1)$. Then $x_n \to 1$ and $[x_n] = 0$, so $f(x_n) \to 0 \neq 1 = f(1)$. Hence [x] is discontinuous at 1.

Marks: statement 1 · sequence 1 · conclusion 1.

4 CALC-24 Q12 · related FYUGP

|x| is continuous everywhere.

KNOW FIRST

T4 ε–δ proofs: x², polynomials, 1/x, |x| M1 Inequality gym

By the reverse triangle inequality, $\big||x| - |c|\big| \leq |x - c|$ for all x, c. Let $c \in \mathbb{R}$ and ε > 0; take δ = ε. If $|x - c| < \delta$ then $\big||x| - |c|\big| \leq |x - c| < \varepsilon$. So |x| is continuous at c; c was arbitrary.

Marks: inequality 1 · δ 1 · conclusion 1.

5

Continuity on a set; g(x) = x.

KNOW FIRST

T1 The ε–δ definition

f is **continuous on** $B \subseteq A$ if f is continuous at every point of B.

For g(x) = x, any c and ε > 0: take δ = ε. If $|x - c| < \delta$ then $|g(x) - g(c)| = |x - c| < \varepsilon$. So g is continuous at every c, i.e. on $\mathbb{R}$.

6 CALC-24 Q3 · related FYUGP

x^2 ($x \le 2$), $x + 10$ ($x > 2$) at 2.

KNOW FIRST

T5 Where functions break **T3** The sequential criterion

$f(2) = 4$. Left limit: $\lim_{x \to 2^-} x^2 = 4$. Right limit: $\lim_{x \to 2^+} (x + 10) = 12$. Since $12 \neq 4$, f is **not continuous** at 2 (a jump of 8). Sequence form: $x_n = 2 + \frac{1}{n} \to 2$ but $f(x_n) = 12 + \frac{1}{n} \to 12 \neq 4$.

Replacing $x + 10$ by $x + 2$ (value 4 at 2) makes the two pieces meet, and f becomes continuous.

Marks: limits 1.5 · conclusion 0.5 · fix 1.

7 PYQ MAT-22 MCQ 3, 8 · adapted

1/x at 0; sgn at 0; which are continuous on x ≥ 0.

KNOW FIRST

T5 Where functions break T3 The sequential criterion

(a) No: 0 is not in the domain of 1/x, and no value at 0 can help, since $x_n = \frac{1}{n}$ gives $1/x_n = n \to \infty$ (no limit).

(b) $x_n = \frac{1}{n} \to 0$ gives $\operatorname{sgn}(x_n) = 1 \not\to 0 = \operatorname{sgn}(0)$. Left limit −1, right limit 1: a **jump** discontinuity.

(c) Only $\sin x$; $\sin(1/x)$ and $1/x$ are not defined at $0 \in A$. (MCQ answers: A; jump.)

8 PYQ MAT-22 MCQ 9 · adapted

Removable hole; x sin(1/x) at 0.

KNOW FIRST

T7 Filling holes: x sin(1/x) versus sin(1/x) T2 Neighbourhoods, limits and isolated points

For $x \neq 2$, $\frac{x^2 + x - 6}{x - 2} = \frac{(x+3)(x-2)}{x-2} = x + 3$, so the limit at 2 is 5. Defining f(2) = 5 makes f continuous at 2 (a **removable** discontinuity).

$|F(x) - F(0)| = |x \sin(1/x)| \leq |x|$. Given ε, take δ = ε: if $|x| < \delta$ then $|F(x) - F(0)| < \varepsilon$. So F is continuous at 0.

9 PYQ MTS-26 Q2 also MAT-22 MCQ 6

Bounded function; continuous but unbounded.

KNOW FIRST

T8 Bounded functions; the Boundedness Theorem
M3 Negating a statement T4 ε–δ proofs: x², polynomials, 1/x, |x|

$f : A \to \mathbb{R}$ is **bounded on A** if there is $M > 0$ with $|f(x)| \leq M$ for all $x \in A$.

$f(x) = 1/x$ on $A = (0, \infty)$ is continuous (T4). It is unbounded: for any $M > 0$, $x_M = \dfrac{1}{M+1} \in A$ gives $f(x_M) = M + 1 > M$. (Similarly $\dfrac{1}{2x}$, the MCQ's function: continuous, not bounded.)

10 PYQ MTS-22 Q2b also MAT-21 Q13b, Q14b

Each hypothesis of the Boundedness Theorem is needed.

KNOW FIRST

T8 Bounded functions; the Boundedness Theorem
T9 The Maximum–Minimum Theorem

(i) unbounded interval	$f(x) = x$ on $[0, \infty)$: continuous, unbounded.
(ii) not closed	$g(x) = 1/x$ on (0, 1]: continuous, unbounded.
(iii) not continuous	$h(x) = 1/x$ on (0, 1], $h(0) = 1$: defined on [0, 1], unbounded.
no maximum	$f(x) = x$ on (0, 1): sup 1, inf 0, neither attained.

So for MTS-22 Q2: yes, boundedness of the interval is necessary.

11 PYQ MAT-22 MCQ 10 · adapted

Absolute max/min.

KNOW FIRST

T9 The Maximum–Minimum Theorem P5

Supremum and infimum

f has an **absolute maximum** on A if some $x^* \in A$ has $f(x^*) \geq f(x)$ for all $x \in A$; an **absolute minimum** if some $x_* \in A$ has $f(x_*) \leq f(x)$ for all $x \in A$.

$1/x$ on $(0, \infty)$: unbounded above, so no maximum; inf = 0 is never reached, so no minimum. x^2 on [−1, 1]: maximum 1 at x = ±1, minimum 0 at x = 0.

12 PYQ MAT-21 Q17b also MAT-22 MCQ 12, MAT-21 MCQ 7

Roots of $xe^x - 2$ and $6x^3 - 3x^2 + 2$.

KNOW FIRST

T10 Location of Roots (bisection)

(a) $f(x) = xe^x - 2$ is continuous on [0, 1]; $f(0) = -2 < 0$ and $f(1) = e - 2 \approx 0.718 > 0$. By the Location of Roots Theorem there is $c \in (0, 1)$ with $f(c) = 0$ (in fact c ≈ 0.8526).

(b) $q(x) = 6x^3 - 3x^2 + 2$ is continuous; $q(-1) = -7 < 0 < 2 = q(0)$. So q has a zero in $(-1, 0) \subset (-1, 1)$ (≈ −0.5606). MCQ answer: "a zero in (−1, 1)".

13 PYQ MAT-21 Q1

Bolzano's IVT; $c^2 = 2$.

KNOW FIRST

T11 Bolzano's Intermediate Value Theorem

Theorem 5.3.7. Let I be an interval and $f : I \to \mathbb{R}$ continuous on I. If $a, b \in I$ and $k \in \mathbb{R}$ satisfies $f(a) < k < f(b)$, then there exists $c \in I$ between a and b with $f(c) = k$.

$f(x) = x^2$ is continuous on [1, 2], and $f(1) = 1 < 2 < 4 = f(2)$. So some $c \in (1, 2)$ has $c^2 = 2$.

14 PYQ MAT-22 Q1 also MAT-21 Q2, MCQs

Uniform continuity; continuous vs UC.

KNOW FIRST

T13 Uniform continuity M2 The ε–δ game and its picture

f is **uniformly continuous on A** if for each $\varepsilon > 0$ there is $\delta(\varepsilon) > 0$ such that if $x, u \in A$ and $|x - u| < \delta(\varepsilon)$, then $|f(x) - f(u)| < \varepsilon$.

Every uniformly continuous function is continuous (fix u = c: the same δ works at c). Not every continuous function is uniformly continuous: $1/x$ on $(0, \infty)$ is continuous but not UC ($x_n = \frac{1}{n}$, $u_n = \frac{1}{n+1}$ get closer, values stay 1 apart). MCQ answer: "every uniformly continuous function is continuous".

15 PYQ MTS-26 Q3 also MAT-21 Q28a

Lipschitz function.

KNOW FIRST

T16 Lipschitz functions

$f : A \to \mathbb{R}$ is **Lipschitz** on A if there is a constant $K > 0$ with $|f(x) - f(u)| \leq K|x - u|$ for all $x, u \in A$. Geometrically (on an interval): every chord of the graph has slope between −K and K.

$|(3x + 1) - (3u + 1)| = 3|x - u|$, so $3x + 1$ is Lipschitz with K = 3.

16 PYQ MTS-25 Q2 also MAT-22 Q15

UC but not Lipschitz.

KNOW FIRST

T16 Lipschitz functions T15 The Uniform Continuity Theorem

$g(x) = \sqrt{x}$ on I = [0, 2]. It is continuous on a closed bounded interval, so uniformly continuous (Theorem 5.4.3). It is not Lipschitz: given any K > 0 put $x = \dfrac{1}{(K+1)^2} \in I$. Then

$$|g(x) - g(0)| = \frac{1}{K+1} > \frac{K}{(K+1)^2} = K|x - 0|.$$

(E.g. K = 5: $\dfrac{1}{6} \approx 0.167 > \dfrac{5}{36} \approx 0.139$.)

Marks: example 1 · UC 1 · not Lipschitz 1.

17 PYQ MTS-22 MCQ 3 · adapted

f + g Lipschitz; f² need not be.

KNOW FIRST

T16 Lipschitz functions M1 Inequality gym

If $|f(x)-f(u)| \le K_1|x-u|$ and $|g(x)-g(u)| \le K_2|x-u|$, then $|(f+g)(x)-(f+g)(u)| \le (K_1+K_2)|x-u|$. So f + g is Lipschitz.

$f(x)=x$ on $[0,\infty)$ is Lipschitz (K = 1). But $|x^2-u^2| = (x+u)|x-u|$; with $x=n+1$, $u=n$ the ratio is $2n+1$, unbounded. So f^2 is not Lipschitz. (MCQ: the false statement is the one about f^2.)

18 PYQ MTS-22 MCQ 4 · adapted

$1/(1+x^2)$ is UC on $\mathbb{R}$.

KNOW FIRST

T16 Lipschitz functions M1 Inequality gym

$f(x)-f(u) = \dfrac{(1+u^2)-(1+x^2)}{(1+x^2)(1+u^2)} = \dfrac{(u-x)(u+x)}{(1+x^2)(1+u^2)}$. Since $\dfrac{|x|}{1+x^2} \le \dfrac{1}{2}$ and $\dfrac{1}{1+u^2} \le 1$ (and the same with x, u swapped),

$$\frac{|u+x|}{(1+x^2)(1+u^2)} \le \frac{|x|}{1+x^2} + \frac{|u|}{1+u^2} \le 1.$$

So $|f(x)-f(u)| \le |x-u|$: Lipschitz with K = 1, hence uniformly continuous (δ = ε).

19 PYQ MAT-22 Q5 also MAT-21 Q3

Step function.

KNOW FIRST

T21 Step functions and piecewise linear approximation

$s : [a, b] \to \mathbb{R}$ is a **step function** if [a, b] is the union of finitely many nonoverlapping intervals $I_1, \ldots, I_n$ with s constant on each I_k.

The given s: the six intervals [−2, −1), [−1, 0], (0, ½), [½, 1), [1, 3], (3, 4] do not overlap and cover [−2, 4], and s is constant on each. So s is a step function (6 distinct values: 0, 1, ½, 3, −2, 2). $f(x) = x$ on [0, 1] takes infinitely many values, so it is not a step function.

20 PYQ MTS-25 Q3 also MAT-22 Q2

Weierstrass Approximation Theorem.

KNOW FIRST

T22 Weierstrass Approximation Theorem T8 Bounded functions; the Boundedness Theorem

Theorem 5.4.14. Let $I = [a, b]$ and $f : I \to \mathbb{R}$ continuous. If ε > 0 is given, there is a polynomial p_ε with $|f(x) - p_\varepsilon(x)| < \varepsilon$ for all $x \in I$.

For 1/x on (0, 1]: any polynomial p is bounded on [0, 1], say $|p| \le B$. If $|1/x - p(x)| < 1$ on (0, 1] then $1/x < B + 1$ there, false near 0. The interval (0, 1] is not closed.

Section B · Paragraph / problem
6 marks each.

21

State and prove the Sequential Criterion.

KNOW FIRST
T3 The sequential criterion P6 Sequences and their limit theorems M3 Negating a statement

Theorem 5.1.3. $f : A \to \mathbb{R}$ is continuous at $c \in A$ iff for every sequence (x_n) in A converging to c, $(f(x_n))$ converges to f(c).

(⇒) Let $x_n \to c$ in A and ε > 0. Continuity gives δ with $|f(x) - f(c)| < \varepsilon$ for $x \in A$, $|x - c| < \delta$. Since $x_n \to c$ there is K with $|x_n - c| < \delta$ for $n \geq K$. Then $|f(x_n) - f(c)| < \varepsilon$ for $n \geq K$, so $f(x_n) \to f(c)$.

(⇐) Suppose f is not continuous at c. Then there is $\varepsilon_0 > 0$ such that for every δ > 0 some $x \in A$ has $|x - c| < \delta$ and $|f(x) - f(c)| \geq \varepsilon_0$. Taking δ = 1/n gives $x_n \in A$ with $|x_n - c| < 1/n$ and $|f(x_n) - f(c)| \geq \varepsilon_0$. Then $x_n \to c$ but $f(x_n) \not\to f(c)$, so the sequence condition fails. By contrapositive, the condition implies continuity. ∎

Marks: statement 1 · (⇒) 2 · (⇐) negation and x_n 3.

22 PYQ MTS-25 Q1

ε–δ for x²; polynomials continuous.

KNOW FIRST

T4 ε–δ proofs: x², polynomials, 1/x, |x| M1
Inequality gym M4 Scratch work, then proof T3
The sequential criterion

Scratch. $|x^2 - c^2| = |x - c||x + c|$, and if $|x - c| < 1$ then $|x + c| \le |x - c| + 2|c| < 1 + 2|c|$.

Proof. Let ε > 0 and $\delta = \min\{1, \frac{\varepsilon}{1 + 2|c|}\}$. If $|x - c| < \delta$ then $|x^2 - c^2| < (1 + 2|c|) \cdot \frac{\varepsilon}{1 + 2|c|} = \varepsilon$. So x^2 is continuous at every c. (E.g. c = 3, ε = 0.01: δ = 1/700.)

Polynomials. Let $p(x) = a_k x^k + \cdots + a_1 x + a_0$ and $x_n \to c$. By the limit theorems for sums and products, $x_n^j \to c^j$ and $p(x_n) \to p(c)$. By the Sequential Criterion, p is continuous at every c. ∎

Marks: scratch 1 · δ and proof 3 · polynomials 2.

23

1/x continuous on (0, ∞); δ for c = ½, ε = 1/10.

KNOW FIRST

T4 ε–δ proofs: x², polynomials, 1/x, |x| M1
Inequality gym

Let c > 0, ε > 0. $\left|\frac{1}{x} - \frac{1}{c}\right| = \frac{|x - c|}{xc}$. If $|x - c| < c/2$ then $x > c/2$, so $\frac{1}{xc} < \frac{2}{c^2}$. Put $\delta = \min\{\frac{c}{2}, \frac{c^2\varepsilon}{2}\}$. If $x > 0$ and $|x - c| < \delta$:

$$\left|\frac{1}{x}-\frac{1}{c}\right| < \frac{2}{c^2}|x-c| < \frac{2}{c^2}\cdot\frac{c^2\varepsilon}{2} = \varepsilon.$$

So 1/x is continuous at every c > 0.

ANSWER **c = ½, ε = 1/10: δ = min{¼, 1/80} = 1/80.**

Check: x = 0.5125 gives $|1/x - 2| \approx 0.049 < 0.1$.

24 PYQ MTS-25 Q16

Dirichlet's function is discontinuous everywhere.

KNOW FIRST

T6 Dirichlet and Thomae T3 The sequential criterion P4 Rationals, irrationals and Archimedes

f(x) = 1 for rational x, 0 for irrational x.

c rational. $x_n = c + \dfrac{\sqrt{2}}{n}$ is irrational and $x_n \to c$. So $f(x_n) = 0 \to 0 \neq 1 = f(c)$.

b irrational. By density of $\mathbb{Q}$ choose rationals y_n with $|y_n - b| < \dfrac{1}{n}$; then $y_n \to b$ and $f(y_n) = 1 \to 1 \neq 0 = f(b)$.

By the Discontinuity Criterion f is discontinuous at every point of $\mathbb{R}$. ∎

Marks: definition 1 · rational case 2 · irrational case 2 · conclusion 1.

25 PYQ MTS-22 MCQ 1 · adapted

Thomae's function.

KNOW FIRST

T6 Dirichlet and Thomae P4 Rationals, irrationals and Archimedes

On $A = \{x > 0\}$: h(x) = 0 for irrational x; $h(m/n) = 1/n$ for $m, n \in \mathbb{N}$ with no common factor.

Rational a = m/n. Irrationals $x_k \to a$ (density) give $h(x_k) = 0 \to 0 \neq 1/n = h(a)$: discontinuous.

Irrational b. Let ε > 0; by Archimedes pick n_0 with $1/n_0 < \varepsilon$. In $(b-1, b+1)$ there are only finitely many rationals with denominator less than n_0, and b is not one of them. Choose δ > 0 smaller than the distance from b to each (and ≤ 1). If $x \in A$, $|x - b| < \delta$, then x is irrational (h(x) = 0) or a rational with denominator ≥ n_0 ($h(x) \leq 1/n_0$). So $|h(x) - h(b)| < \varepsilon$: continuous at b. ∎ (MCQ: continuous only at irrationals.)

26 PYQ MTS-26 Q16 also MAT-22 Q27, MAT-21 Q13a, MTS-22 Q2a

State and prove the Boundedness Theorem.

KNOW FIRST

T8 Bounded functions; the Boundedness Theorem P7 Subsequences and Bolzano–Weierstrass T3 The sequential criterion M5 Contradiction, and "one point for each n"

Theorem 5.3.2. Let I = [a, b] be a closed bounded interval and $f : I \to \mathbb{R}$ continuous on I. Then f is

bounded on I.

1. Suppose not. Then for each n ∈ ℕ there is $x_n \in I$ with $|f(x_n)| > n$.
   Negation of boundedness with M = n.

2. (x_n) is bounded (I is bounded), so by Bolzano–Weierstrass a subsequence $x_{n_r} \to x$.
   BW 3.4.8.

3. Since $a \le x_{n_r} \le b$, also $a \le x \le b$, so $x \in I$.
   I is closed (3.2.6).

4. f is continuous at x, so $f(x_{n_r}) \to f(x)$; hence $(f(x_{n_r}))$ is bounded.
   Sequential criterion; convergent ⇒ bounded (3.2.2).

5. But $|f(x_{n_r})| > n_r \ge r$ for every r: unbounded. Contradiction, so f is bounded. ∎
   $n_r \ge r$.

Marks: statement 1 · negation 1 · BW 1 · x ∈ I 1 · continuity 1 · contradiction 1.

27 PYQ MAT-22 Q13

inf ≤ k ≤ sup ⇒ k is a value; $x^3 - 3x$, k = 1.

KNOW FIRST

T11 Bolzano's Intermediate Value Theorem T9 The Maximum–Minimum Theorem

By the Max–Min Theorem there are $c_*, c^* \in I$ with $f(c_*) = \inf f(I)$ and $f(c^*) = \sup f(I)$. If k equals one of these, take c = c_* or c^*. Otherwise $f(c_*) < k < f(c^*)$, and Bolzano's theorem on the interval between c_* and c^* gives c with $f(c) = k$. ∎

Illustration. $f(x) = x^3 - 3x$ on [−1.5, 1.5]: inf = −2 (at 1), sup = 2 (at −1), so k = 1 is taken. Indeed $c^3 - 3c = 1$ has the root $c = 2\cos 100^\circ \approx -0.347$ in I.

28 PYQ MAT-21 Q15

f([a, b]) is a closed bounded interval.

KNOW FIRST

T12 Images of intervals T9 The Maximum–Minimum Theorem T11 Bolzano's Intermediate Value Theorem

Theorem 5.3.9. If I is a closed bounded interval and $f : I \to \mathbb{R}$ is continuous, then f(I) is a closed bounded interval.

Proof. Let $m = \inf f(I)$, $M = \sup f(I)$ (they exist: f is bounded by 5.3.2). By Max–Min both are values of f, and every value lies between them: $f(I) \subseteq [m, M]$. By Corollary 5.3.8 every $k \in [m, M]$ is a value: $[m, M] \subseteq f(I)$. So $f(I) = [m, M]$. ∎

Example. $f(x) = x^3 - 3x$ on [−1.5, 1.5]: f(a) = 1.125, f(b) = −1.125, but $f(I) = [-2, 2]$ (max 2 at −1, min −2 at 1), not $[f(b), f(a)] = [-1.125, 1.125]$.

29 PYQ MTS-25 Q17 also MAT-22 Q14

Preservation of Intervals.

KNOW FIRST

T12 Images of intervals T11 Bolzano's Intermediate Value Theorem P10 Nested intervals and what an interval is

Theorem 5.3.10. Let I be an interval and $f : I \to \mathbb{R}$ continuous on I. Then f(I) is an interval.

Proof. Let $\alpha < \beta$ be in f(I), say $\alpha = f(a)$, $\beta = f(b)$ with $a, b \in I$. If $\alpha < k < \beta$, Bolzano's theorem gives c between a and b (so c ∈ I) with $f(c) = k$, i.e. $k \in f(I)$. Thus $[\alpha, \beta] \subseteq f(I)$ for all $\alpha < \beta$ in f(I), and by the characterisation 2.5.1, f(I) is an interval (a single point if f is constant). ∎

Images. For $f(x) = \dfrac{1}{x^2+1}$: $f((-1,1)) = (\dfrac{1}{2}, 1]$ and $f([0,\infty)) = (0,1]$. Intervals, but "open" and "closed" are not preserved.

Marks: statement 1 · proof 3 · two images 2.

30 PYQ MAT-21 Q16 also MTS-22 MCQ 2

Nonuniform Continuity Criteria; 1/x.

KNOW FIRST

T14 Proving "not uniformly continuous" T16 Lipschitz functions M6 Choosing sequences to break things

5.4.2. For $f : A \to \mathbb{R}$ the following are equivalent: (i) f is not uniformly continuous on A; (ii) there is $\varepsilon_0 > 0$ such that for every δ > 0 there are $x_\delta, u_\delta \in A$ with $|x_\delta - u_\delta| < \delta$ and $|f(x_\delta) - f(u_\delta)| \geq \varepsilon_0$; (iii) there are $\varepsilon_0 > 0$ and sequences $(x_n), (u_n)$ in A with $\lim(x_n - u_n) = 0$ and $|f(x_n) - f(u_n)| \geq \varepsilon_0$ for all n.

1/x on (0, ∞). $x_n = \dfrac{1}{n}$, $u_n = \dfrac{1}{n+1}$: $x_n - u_n = \dfrac{1}{n(n+1)} \to 0$ and $|f(x_n) - f(u_n)| = |n - (n+1)| = 1$. With $\varepsilon_0 = 1$, (iii) holds: not uniformly continuous.

On [a, 1), a > 0. $\left|\frac{1}{x}-\frac{1}{u}\right| = \frac{|x-u|}{xu} \le \frac{1}{a^2}|x-u|$: Lipschitz with $K = 1/a^2$, hence uniformly continuous (δ = εa²). (MCQ: uniformly continuous on [a, 1).)

31 PYQ MAT-21 Q28b

Lipschitz ⇒ uniformly continuous; sin x.

KNOW FIRST

T16 Lipschitz functions T13 Uniform continuity

Theorem 5.4.5. If f is Lipschitz on A with constant K, it is uniformly continuous on A.

Proof. Given ε > 0 let δ = ε/K. If $x, u \in A$ and $|x-u| < \delta$, then $|f(x)-f(u)| \le K|x-u| < K\delta = \varepsilon$. δ depends only on ε (and the fixed K). ∎

sin x. $|\sin x - \sin u| = 2\left|\cos\frac{x+u}{2}\right|\left|\sin\frac{x-u}{2}\right| \le 2 \cdot 1 \cdot \frac{|x-u|}{2} = |x-u|$ (using $|\sin t| \le |t|$). So K = 1 and δ = ε.

32 PYQ MTS-26 Q24b also MAT-22 Q36b

√x is uniformly continuous on [0, ∞).

KNOW FIRST

T17 √x is uniformly continuous on [0, ∞) T15 The Uniform Continuity Theorem T16 Lipschitz functions

On I = [0, 2] (MAT-22 Q36b): $\sqrt{x}$ is continuous on a closed bounded interval, so uniformly continuous (5.4.3); call its δ $\delta_I(\varepsilon)$.

On J = [1, ∞): $|\sqrt{x}-\sqrt{u}| = \dfrac{|x-u|}{\sqrt{x}+\sqrt{u}} \le \dfrac{1}{2}|x-u|$, Lipschitz with $K = \dfrac{1}{2}$; $\delta_J = 2\varepsilon$.

Together. Let $\delta = \min\{1, \delta_I(\varepsilon), \delta_J(\varepsilon)\}$. If $x, u \ge 0$ and $|x-u| < \delta$: if both are ≥ 1 they lie in J; otherwise one is < 1 and the other < 2, so both lie in I. Either way $|\sqrt{x}-\sqrt{u}| < \varepsilon$. ∎

Direct alternative. $|\sqrt{x}-\sqrt{u}| \le \sqrt{|x-u|}$, so δ = ε² works everywhere.

33 PYQ MAT-21 Q22

Sum of UC functions is UC; product need not be.

KNOW FIRST

T18 Sums and products of uniformly continuous functions T14 Proving "not uniformly continuous"

Sum. Given ε, take $\delta = \min\{\delta_f(\varepsilon/2), \delta_g(\varepsilon/2)\}$. If $|x-u| < \delta$: $|(f+g)(x) - (f+g)(u)| \le |f(x) - f(u)| + |g(x) - g(u)| < \dfrac{\varepsilon}{2} + \dfrac{\varepsilon}{2} = \varepsilon$.

Product. $f(x) = g(x) = x$ are uniformly continuous on $\mathbb{R}$ (δ = ε). Their product x^2 is not: $x_n = n$, $u_n = n + \dfrac{1}{n}$ give $x_n - u_n \to 0$ but $|x_n^2 - u_n^2| = 2 + \dfrac{1}{n^2} \ge 2$ (5.4.2 with $\varepsilon_0 = 2$).

34 PYQ MAT-21 Q36a also MTS-22 Q3

UC maps Cauchy sequences to Cauchy sequences.

KNOW FIRST

T19 Uniform continuity keeps Cauchy sequences Cauchy P8 Cauchy sequences

Theorem 5.4.7. Let (x_n) be Cauchy in A and ε > 0. By uniform continuity choose δ with $|f(x) - f(u)| < \varepsilon$ whenever $x, u \in A$, $|x - u| < \delta$. Since (x_n) is Cauchy there is H with $|x_n - x_m| < \delta$ for $n, m > H$. Then $|f(x_n) - f(x_m)| < \varepsilon$ for $n, m > H$, so $(f(x_n))$ is Cauchy. ∎

Fails for mere continuity: $f(x) = 1/x$ on (0, 1) and $x_n = \dfrac{1}{n+1}$ (Cauchy, → 0): $f(x_n) = n + 1$ is unbounded, so not Cauchy.

35 PYQ MTS-26 Q17

Step-function approximation; f(x) = x, ε = 0.1.

KNOW FIRST

T21 Step functions and piecewise linear approximation T15 The Uniform Continuity Theorem

Theorem 5.4.10. f is uniformly continuous on [a, b] (5.4.3); let δ = δ(ε). Choose m ∈ ℕ with $h = \dfrac{b-a}{m} < \delta$. Let $I_1 = [a, a + h]$, $I_k = (a + (k - 1)h, a + kh]$ for k = 2, ..., m. Define $s_\varepsilon(x) = f(a + kh)$ for $x \in I_k$. This is a step function, and for $x \in I_k$, $|x - (a + kh)| \le h < \delta$, so $|f(x) - s_\varepsilon(x)| < \varepsilon$. ∎

Example. f(x) = x: $|f(x) - f(u)| = |x - u|$, so δ = 0.1. We need h < 0.1 strictly: m = 11, h = 1/11. $s_\varepsilon = \dfrac{k}{11}$ on I_k, with error at most $\dfrac{1}{11} \approx 0.091 < 0.1$. (m = 10 fails at x = 0, where the error is exactly 0.1.)

36

Piecewise linear approximation; x² with four pieces.

KNOW FIRST

T21 Step functions and piecewise linear approximation T15 The Uniform Continuity Theorem

Definition. $g : [a, b] \to \mathbb{R}$ is piecewise linear if [a, b] is a union of finitely many disjoint intervals on each of which g is linear.

Proof. Choose δ from uniform continuity and nodes $x_k = a + kh$ with $h < \delta$. Let g_ε be the chord through $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$ on each $[x_{k-1}, x_k]$; it is continuous since neighbouring chords meet at the nodes. For $x = (1 - \lambda)x_{k-1} + \lambda x_k$ $(0 \le \lambda \le 1)$,

$f(x) - g_\varepsilon(x) = (1 - \lambda)[f(x) - f(x_{k-1})] + \lambda[f(x) - f($

and both brackets are less than ε in size, so $|f(x) - g_\varepsilon(x)| < \varepsilon$. ∎

Example. On a piece $[p, p + h]$ the chord of x^2 minus x^2 is $(x - p)(p + h - x)$, largest at the midpoint: $h^2/4$. With h = ¼:

ANSWER largest error $= \dfrac{1}{64} \approx 0.0156$.

Section C · Essays

10 marks each.

37 PYQ MTS-25 Q24 also MAT-21 Q17a

State and prove the Location of Roots Theorem.

KNOW FIRST

T10 Location of Roots (bisection) P10 Nested intervals and what an interval is T3 The sequential criterion P6 Sequences and their limit theorems M7 Bisection

Theorem 5.3.5. Let I = [a, b] and let $f : I \to \mathbb{R}$ be continuous on I. If $f(a) < 0 < f(b)$, or if $f(a) > 0 > f(b)$, then there exists $c \in (a, b)$ such that $f(c) = 0$.

Proof (case $f(a) < 0 < f(b)$).

1. Let $I_1 = [a_1, b_1] = [a, b]$ and $p_1 = \dfrac{a_1 + b_1}{2}$. If $f(p_1) = 0$, take $c = p_1$. If $f(p_1) > 0$ put $I_2 = [a_1, p_1]$; if $f(p_1) < 0$ put $I_2 = [p_1, b_1]$. In both cases $f(a_2) < 0 < f(b_2)$.
   Bisection keeps the sign change.

2. Continue. Either we reach some p_n with $f(p_n) = 0$, or we get nested intervals $I_n = [a_n, b_n]$ with $f(a_n) < 0 < f(b_n)$ and $b_n - a_n = \dfrac{b - a}{2^{n-1}}$.
   Induction; each step halves the length.

3. By the Nested Intervals Property there is c in every I_n. Since $a_n \le c \le b_n$ and $b_n - a_n \to 0$, we get $a_n \to c$ and $b_n \to c$.
   2.5.2; $|a_n - c| \le b_n - a_n$.

4. f is continuous at $c \in I$, so $f(a_n) \to f(c)$ and $f(b_n) \to f(c)$.
   Sequential criterion 5.1.3.

5. $f(a_n) < 0$ for all n gives $f(c) \le 0$; $f(b_n) > 0$ gives $f(c) \ge 0$. Hence $f(c) = 0$.
   Order theorem 3.2.6.

6. Since $f(a) < 0$ and $f(b) > 0$, c is neither a nor b: $c \in (a, b)$.
   f(c) = 0.

7. If $f(a) > 0 > f(b)$, apply the above to $g = -f$, which is continuous with $g(a) < 0 < g(b)$. ∎
   A zero of –f is a zero of f.

Error estimate: the midpoint p_n is within $(b-a)/2^n$ of c.

Marks: statement 2 · construction 3 · nested intervals and limits 2 · continuity and sign 2 · conclusion 1.

38 PYQ MAT-22 Q36a also MAT-21 Q14a, MAT-22 MCQ 11

State and prove the Maximum–Minimum Theorem.

KNOW FIRST

T9 The Maximum–Minimum Theorem T8 Bounded functions; the Boundedness Theorem P5 Supremum and infimum P7 Subsequences and Bolzano–Weierstrass T3 The sequential criterion

Theorem 5.3.4. Let I = [a, b] be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Then f has an absolute maximum and an absolute minimum on I. (MCQ: this needs I closed and bounded.)

Lemma used. Boundedness Theorem 5.3.2: f is bounded on I.

1. $f(I) = \{f(x) : x \in I\}$ is non-empty and bounded, so $s^* = \sup f(I)$ and $s_* = \inf f(I)$ exist.
   5.3.2 and Completeness.

2. For each n, $s^* - \dfrac{1}{n}$ is not an upper bound of f(I): choose $x_n \in I$ with $s^* - \dfrac{1}{n} < f(x_n) \leq s^*$.
   Definition of supremum.

3. (x_n) is bounded, so by Bolzano–Weierstrass some subsequence $x_{n_r} \to x^*$; since $a \leq x_{n_r} \leq b$, $x^* \in I$.
   I bounded and closed.

4. f is continuous at x^*: $f(x_{n_r}) \to f(x^*)$.
   Sequential criterion.

5. From $s^* - \dfrac{1}{n_r} < f(x_{n_r}) \leq s^*$, the Squeeze Theorem gives $f(x_{n_r}) \to s^*$.
   $1/n_r \to 0$.

6. By uniqueness of limits $f(x^*) = s^*$, so $f(x^*) \geq f(x)$ for all $x \in I$: an absolute maximum at x^*.
   s^* is an upper bound.

7. For the minimum, choose x_n with $s_* \le f(x_n) < s_* + \dfrac{1}{n}$ and repeat steps 3–6 (or apply the result to −f). ∎
   Symmetry.

Hypotheses. Not closed: $f(x) = x$ on (0, 1) has no max or min. Not bounded: x on [0, ∞) has no max. Not continuous: $h(x) = x$ on [0, 1), $h(1) = 0$ has no max.

Marks: statement 1 · sup exists 1 · approximating x_n 2 · BW and $x^* \in I$ 2 · continuity and squeeze 2 · max 1 · min 1.

39 PYQ MTS-26 Q24a also MAT-22 Q28, MAT-21 Q27

State and prove the Uniform Continuity Theorem.

KNOW FIRST
T15 The Uniform Continuity Theorem T14 Proving "not uniformly continuous" P7 Subsequences and Bolzano–Weierstrass T3 The sequential criterion

Theorem 5.4.3. Let I be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Then f is uniformly continuous on I.

1. Suppose f is not uniformly continuous on I. By 5.4.2 there are $\varepsilon_0 > 0$ and sequences $(x_n), (u_n)$ in I with $|x_n - u_n| < \dfrac{1}{n}$ and $|f(x_n) - f(u_n)| \ge \varepsilon_0$ for all n.
   Nonuniform Continuity Criteria with δ = 1/n.

2. (x_n) is bounded, so by Bolzano–Weierstrass a subsequence $x_{n_k} \to z$. Since I is closed, $z \in I$.
   I bounded and closed.

3. $|u_{n_k} - z| \le |u_{n_k} - x_{n_k}| + |x_{n_k} - z| < \frac{1}{n_k} + |x_{n_k} - z| \to 0$, so $u_{n_k} \to z$.
   Triangle inequality.

4. f is continuous at z, so $f(x_{n_k}) \to f(z)$ and $f(u_{n_k}) \to f(z)$.
   Sequential criterion.

5. Hence $|f(x_{n_k}) - f(u_{n_k})| \to |f(z) - f(z)| = 0$, contradicting $|f(x_{n_k}) - f(u_{n_k})| \ge \varepsilon_0$ for all k.
   Limit of a difference.

6. Therefore f is uniformly continuous on I. ∎

Hypotheses. 1/x on (0, 1] (not closed) and x² on [0, ∞) (not bounded) are continuous but not uniformly continuous.

Marks: statement 1 · negation and sequences 3 · BW and z ∈ I 2 · second sequence 2 · continuity and contradiction 2.

40 PYQ MAT-22 Q29

State and prove the Continuous Extension Theorem.

KNOW FIRST

T20 The Continuous Extension Theorem T19 Uniform continuity keeps Cauchy sequences Cauchy T15 The Uniform Continuity Theorem P8 Cauchy sequences P9 Limits of functions T7 Filling holes: x sin(1/x) versus sin(1/x)

Theorem 5.4.8. A function f is uniformly continuous on the interval (a, b) if and only if it can be defined at the endpoints a and b such that the extended function is continuous on [a, b].

(⇐) If the extension is continuous on [a, b], it is uniformly continuous there (Theorem 5.4.3), and the same δ works on the subset (a, b).

(⇒) We show $\lim_{x\to a} f(x)$ exists; the end b is similar.

1. Let (x_n) in (a, b) converge to a. It is a Cauchy sequence.
   Convergent ⇒ Cauchy (3.5.3).

2. By Theorem 5.4.7, $(f(x_n))$ is Cauchy, hence converges to some L.
   Uniform continuity; Cauchy ⇒ convergent (3.5.5).

3. If (u_n) is another sequence in (a, b) converging to a, then $u_n - x_n \to 0$, so by uniform continuity $f(u_n) - f(x_n) \to 0$, hence $f(u_n) \to L$.
   Given ε, take δ; eventually $|u_n - x_n| < \delta$.

4. Every sequence tending to a gives the same limit L, so $\lim_{x\to a} f(x) = L$ by the sequential criterion for limits (4.1.8). Define $f(a) := L$.
   P9.

5. Similarly define f(b). The extended f is continuous at a and b (limit = value), and at every interior point as before; so it is continuous on [a, b]. ∎
   Remark 5.1.7.

Use. $\sin(1/x)$: no limit at 0, so not UC on (0, 1); $x\sin(1/x)$: limit 0, so UC.

Marks: statement 2 · (⇐) 2 · Cauchy image 2 · independence of sequence 2 · definition at ends and continuity 2.